

HOLOGRAPHY

INTRODUCTION

Conventional photography gives a two dimensional record of the intensity distribution from a three dimensional object. All information on the relative phases of light waves, originating from different points on the object, is lost. In holography, (holos = whole) both the amplitude and the **phase** of the light scattered by an object is recorded i.e. a hologram contains enough information to reconstruct a three dimensional image of the original object.

Since light recording media are only sensitive to changes in light **intensity**, the phase information is converted to intensity variations by employing coherent illumination and adding a **reference wave** (plane or spherical), derived from the same light source, to the waves scattered by the object. Thus, a hologram is a record of the interference pattern formed by a reference beam and the light waves scattered from the object. Interference pattern spacings are $\sim \lambda/2$ so that the resolution of the hologram recording medium must be of the same order. When using a HeNe laser as the source $\lambda/2 = 0.3 \mu\text{m}$ and a resolution of 3000 lines/mm is required (200 lines/mm is normally ample for conventional photography).

Holograms are classified according to:

- II the thickness of the recording material
- II the type of modulation imposed on the illuminating (readout) wave
- II their recording geometry
- II the mode of image formation

A **thin** (plane) hologram is one where the thickness of the recording medium is smaller than the fringe spacing. In **thick** (volume) holograms the thickness is greater than the fringe spacing.

With **amplitude** holograms the interference pattern is recorded as an optical density variation of the medium, which leads to absorption (amplitude modulation) of parts of the illuminating wave. In **phase** holograms the refractive index of the recording medium varies with position, imposing a phase modulation onto the illuminating beam.

A **Fresnel** hologram is obtained when the object is sufficiently close to the recording medium so that the Fresnel diffraction pattern of the object is recorded. If the distance is large enough to record the far field pattern, a **Fraunhofer** hologram results. A **Fourier transform** hologram is produced when a lens, one focal length distant from both the object and the recording medium, is used to give the far field pattern of the object. In this last case the reference beam must be a plane wave. If the reference beam originates from a point the same distance from the recording medium as the object, a **lenseless Fourier transform** hologram is formed.

An **in-line** hologram is obtained when the two beams, (object and reference) interfering at the recording medium, are travelling in nearly the same direction. If their

directions are substantially different, an **off-axis** hologram results while, if the two beams are travelling in essentially opposite directions a **reflection** hologram is formed. The image, in the case of off-axis and in-line holograms, is formed by passing a readout beam through the hologram so that these are known as **transmission** holograms. In the case of reflection holograms, an image is formed when the readout beam is **reflected** from the hologram.

In this experiment you will produce thick, phase, Fresnel holograms, both transmission (off-axis) and reflection. Ref. 1 gives a good general introduction while Ref. 2 explains how reflection and phase holograms work. You will also try a simple **double-exposure holographic interferometry** experiment (qualitative only). Refs. 1 and 4 describe this technique.

WARNINGS !

SAFETY GOGGLES MUST BE WORN DURING THIS EXPERIMENT

- II The laser used is a 10 mW HeNe laser whose direct beam, or a specular reflection, can **severely damage** your retina. Use the **beam attenuator** when aligning optical components. Close the **beam shutter** on the laser when not using it or when moving optical components around on the table, screwing them down etc. Keep the table uncluttered - remove any unnecessary objects. Be aware of the position of **other people** in the room - don't damage their eyes with **stray reflections**. Keep the **room lights on** whenever possible.
- II Some of the optical components used (e.g. spatial filter, front surface mirrors, beamsplitter) are particularly expensive. Handle all the components with care; carefully lay them down on the table if their bases are not screwed down. **Do not try to wipe or clean** the beam splitter or the mirrors and keep your fingers off the reflecting surfaces.
- II Open the box of holographic plates only in **complete darkness**.
- II Wear a **lab coat and gloves**, and turn the **exhaust fan** on, when using chemicals in the dark room.

EXPERIMENTAL

Expt. 1 The Michelson Interferometer

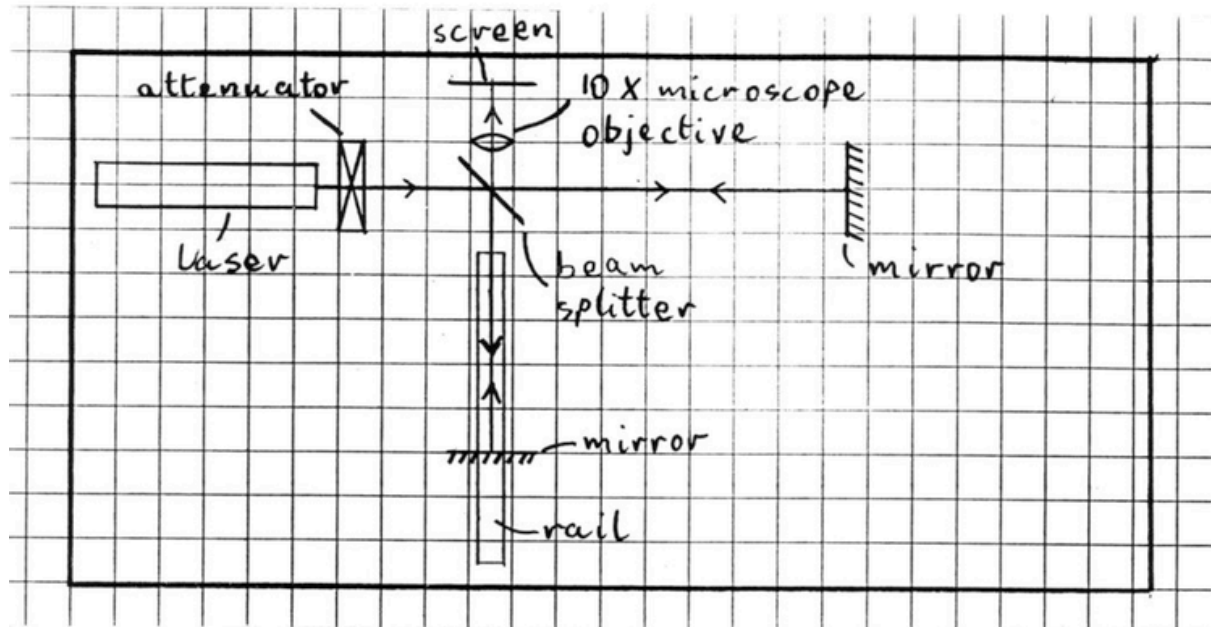


Figure 1. Michelson interferometer set-up (1div =10cm)

Using the slotted indicator, check that the laser is level (i.e. the beam path is parallel to the holography table) and, using Fig.1 as a guide, set up a Michelson interferometer to give about 4-6 widely spaced fringes on the screen. Study the stability of the fringes with various vibrations e.g. stamp your foot, stir up air currents, knock on the table, walls etc. and comment on how serious each type of disturbance would be in the recording of a hologram. Check the effect of vibrations due to opening the shutter.

Investigate fringe visibility as a function of optical path difference between the interferometer arms and determine (roughly) the coherence length of the laser.

Question 1: *How does the coherence length of the laser limit the size of an object which can be holographed?*

Expt. 2 Transmission Hologram

N.B. The laser should be switched on at least 2 hours, and the air cleaner at least 1 hour, before any holograms are exposed. Check that the laser beam is level and vertically polarised and use the **beam attenuator** when setting up.

Question 2: *Why is it preferable to have the beam polarisation perpendicular to the beam's plane?*

Use Fig. 2 as a guide to setting up. Mount the plate holder and the object (one of the white painted (why?) statuettes) at the correct height and 15 - 20 cm from the plate. Insert a white card in the plate holder and tilt the reference mirror so that the reflected beam strikes the card's centre. Using a string, measure the distance from the beam splitter, via the reference mirror, to the centre of the plate holder (**don't touch the splitter or mirror surfaces!**). Set up the object mirror so that the distance $\text{splitter} - \text{mirror} - \text{object} - \text{plate}$ is the same as the distance measured above and tilt the mirror so that the reflected beam is level and strikes the object centrally.

Question 3: Why should the object and reference paths be equal?

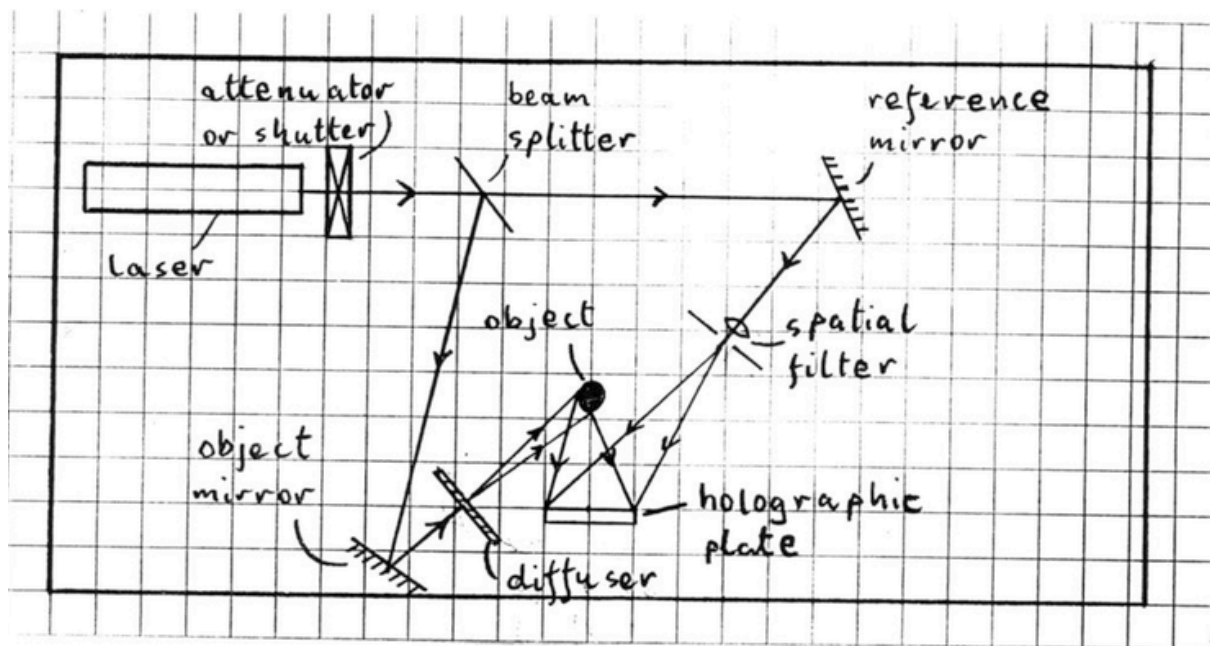


Figure 2. Transmission holography set-up (1div =10cm)

Remove the pinhole from the spatial filter and position the filter in the reference beam so that the filter lens is 40 - 45 cm from the plate holder. Adjust the filter so that the holder is centrally illuminated (use card in holder) then insert the pinhole and adjust it to give the maximum illumination - this is easier said than done; a white card, perseverance and patience are all necessary. You will need to adjust the beam splitter to give maximum intensity in the reference beam and you may need to remove the beam attenuator (take care if you do!). Make sure that the object does not obstruct any part of the reference beam from striking the plate

Question 4: Why are lasers emitting in the TEM_{00} mode preferred for holography?

Now position the diffuser to evenly illuminate the object.

Block the reference beam, remove the beam attenuator, switch off the lights and look at the object **through the plate holder** (don't look down the reference or object beams!), what you see is what the final image will look like, so readjust the object position at this stage if you wish.

Switch off the air filter.

Use the power meter to measure the laser light intensity in the **plane of the holographic plate**. Measure the intensity, of both the reference and object beams, in about 6 to 9 positions in this plane, calculate the averages and find the ratio of reference/object beam intensities. Adjust the beam splitter to give $\sim 4/1$ reference/object intensity then calculate the exposure time required (check with a demonstrator).

Using a white card, monitor the illumination on the plate holder and place baffle screens in strategic positions so that the plate does not **>see=** any scattered light - it should see **only** the reference beam and the light scattered from the object.

Place the shutter in position, set the correct exposure time on the timer and practise actuating the shutter with the cable release. Activate the compressed gas vibration isolator system for the holography table (check with a demonstrator) and...

You are at last ready to expose the plate!

Check that the warning light is on above the outside of the door, lock the door, close the shutter and remove the attenuator. Switch off the room lights and, in complete darkness insert the holographic plate into the holder with the **emulsion side towards the object** (the emulsion is on the side nearest the box label, or use the **>sticky=** test to find it). Stand perfectly still and allow everything to settle (table, air currents etc.) for at least one minute then open the shutter for the required time and close it. Slide the plate across to allow exposure of the other half and again allow the surroundings to settle before exposing.

Now remove the plate from the holder and place in the transport container. Make sure the transport container and **the box of unexposed plates** are both tightly closed before switching the room lights on again. Process the plate as soon as possible (the latent image slowly fades i.e. the deposited silver slowly converts to a halide again) using the **direct** processing technique (information folder) to obtain a phase hologram.

Question 5: *What is diffraction efficiency? Why do phase holograms normally give the brightest images?*

The dry hologram may be viewed by inserting it in the plate holder (emulsion side as when exposed) and shining only the reference beam through it. If you look through the hologram (don't look down the beam!), the image should appear in the same position where the object was - this is the **virtual** image.

Also try to view the **real** image and to project it on a card.

Question 6: *How does the hologram form a virtual **and** a real image ?*

Try to view the virtual image using a sodium or a mercury lamp. Describe and explain what you see

Expt. 3 Double-exposure holographic interferometry

Replace the statuette with the white painted can, readjust the light ratio to 4:1 and take an exposure lasting half the calculated exposure time. Leave the plate in the holder and cover it. Place a weight on top of the can to stress it (be careful not to move the can in any other way) and on the **same** plate take another exposure of the stressed can, again for half the calculated time.

Process and view the hologram as before. Describe and explain what you see (Ref. 1).

Expt. 4 Reflection Hologram

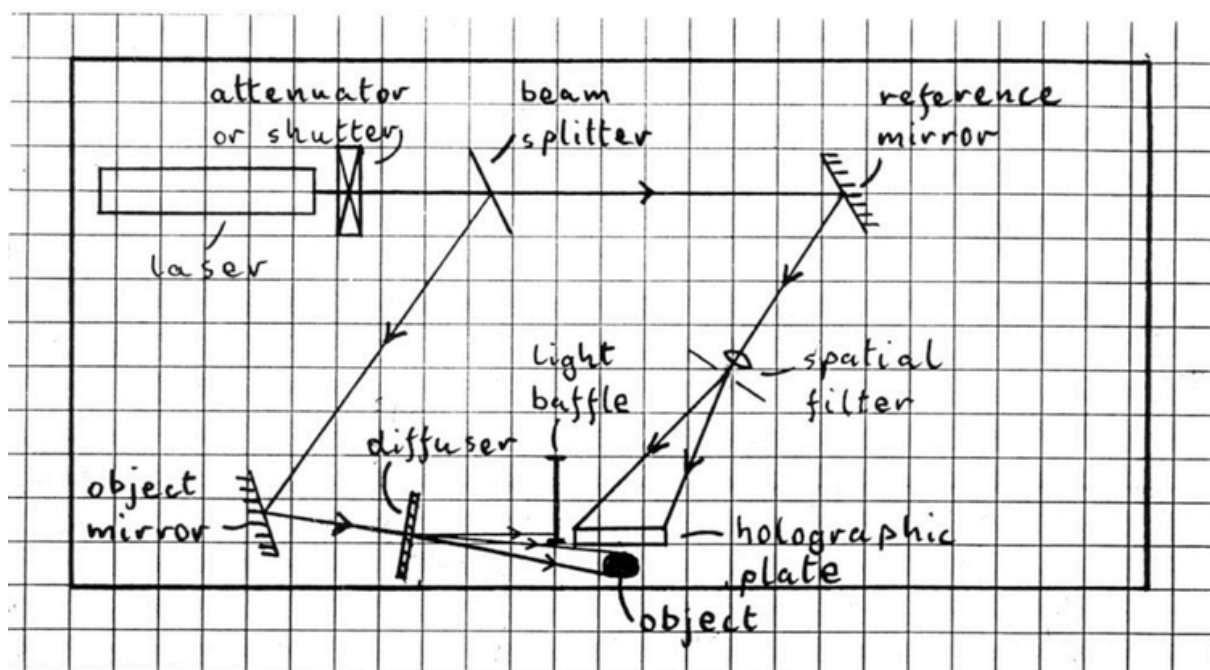


Figure 3. Reflection holography set-up (1div =10cm)

Using Fig. 3 as a guide, set up the table for reflection holography. Use a statuette as the object and position it as close as possible to the plate. Follow the method outlined in Expt. 2 for setting up the various optical components. Again, look **through** the plate holder at the object lit up by the object beam, to see what the final image will look like. Adjust the reference/object beam ratio to $\sim 2/1$. The emulsion side of the plate should be **towards the object**.

Take an exposure and use the **reversal** processing technique to obtain a phase hologram. The hologram may be viewed in sunlight or by using a 'point' source of white light such as a projector lamp or other small filament unfrosted bulb e.g. a 'penlite' torch. The illumination should come from the non-emulsion side, at approximately the same angle as the reference beam during exposure. The viewer needs to be on the same side of the plate as the light source.

What is the colour of the image you observe? Why? Slowly rotate the plate about an axis perpendicular to the illumination direction and observe the colour of the image. Explain what you observe. Use a 'penlite' torch to view the hologram. What happens to the size of the image if you bring the torch close to the plate? Explain.

Turn the plate around so that the emulsion side is illuminated and find the real image (you will need to tilt the plate around a bit). Describe and explain what you see.

Try to view an image by transmitting white light through the plate (it may be too faint to see) and describe what you see.

OPTIONAL EXPERIMENTS (see a demonstrator)

- Make a reflection hologram of an object of your own choice (objects should have pronounced 3D features, be non-shiny and white, red or orange. The object must be securely mounted to minimise vibrations.
- Make a double exposure **reflection** hologram of an unstressed/stressed object.
- Make a white light hologram without using the beam splitter (a Denisyuk hologram).
- Try making an unbleached (i.e. absorption) transmission hologram.

REFERENCES

- 1) O'Shea D.C., Callen W.R. and Rhodes W.T., **An Introduction to Lasers and their Applications** (1978) pp. 198 - 209
- 2) Kasper J.E. and Feller S.A., **The Complete Book of Holograms** (1987)
- 3) Hariharan P., Phot. Sci. and Eng. **24**, 105 (1980)
- 4) Weingartner I. J., Phys. **E16**, 16 (1983)